

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)	(i)		Equation with displayed or structural formulae e.g. $\text{CH}_3\text{CH}(\text{CH}_3)\text{CHClCH}_2\text{CH}_3 + \text{NaOH} \rightarrow \text{CH}_3\text{CH}(\text{CH}_3)\text{CHOHCH}_2\text{CH}_3 + \text{NaCl}$ $\text{CH}_3\text{CH}(\text{CH}_3)\text{CHClCH}_2\text{CH}_3 + \text{OH}^- \rightarrow \text{CH}_3\text{CH}(\text{CH}_3)\text{CHOHCH}_2\text{CH}_3 + \text{Cl}^-$		1		1		
		(ii)		Nucleophilic substitution	1			1		
		(iii)		Neutralise hydroxide with nitric acid and add aqueous silver nitrate (1) White precipitate forms (1) Accept heat with acidified dichromate (1) orange to green (1)	2			2		2
	(b)	(i)		Rate $\propto$ concentration of halogenoalkane with explanation e.g. concentration doubles, rate doubles (1) Rate not affected by concentration of OH^- (1)			2	2	1	
		(ii)		Rate would be faster because C—X bond needs to be broken (1) C—Br is weaker than C—Cl / takes less energy to break (1) This outweighs effect of greater dipole in C—Cl / chlorine being more electronegative (1)	3			3		
				Question 10 total	6	1	2	9	1	2